

Comparative Analysis of Machine Learning and Reinforcement Learning for Stock Prediction on various U.S. Markets

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1 Background and Research Question

Reinforcement learning (RL) has emerged as an alternative, framing trading as a sequential decision-making problem. Moody and Saffell (2001) introduced direct reinforcement methods for trading, showing that RL can learn profitable strategies by optimizing a reward function tied to returns [6]. More recent studies highlight deep RL approaches, such as deep Q-networks and policy gradient methods, which can adaptively manage trading actions under uncertainty [5, 4, 2]. Despite these advances, research comparing supervised ML and RL across multiple stock markets remains limited, motivating the current study.

Overall, the literature indicates that ML methods excel at short-term predictive accuracy, while RL offers the potential for direct optimization of trading performance. Integrating insights from both approaches could lead to improved profitability and risk management in stock trading applications.

1.1 Sub-question 1: Predictive Performance

Research Question: How well do different learning methods identify stock market patterns over short and long time periods? We will compare supervised models (regression, random forests, LSTMs) with reinforcement learning approaches. The focus is on predictive performance and directional accuracy.

Hypothesis: RL methods may initially struggle due to overfitting and unstable training. With well-designed reward functions, RL may perform as well or better than supervised models over longer periods [6].

1.2 Sub-question 2: Risk Management

Research Question: Can trial-and-error learning methods (RL) manage risk better than traditional prediction models? RL might naturally learn strategies like cutting losses or adjusting trade sizes without extra rules.

Hypothesis: RL methods may face challenges in controlling risk effectively because of overfitting and instability during training, potentially leading to higher volatility and larger drawdowns than supervised models [5, 4, 2].

1.3 Sub-question 3: Accuracy, Profitability, and Risk Trade-off

Research Question: What is the relationship between accuracy, profitability, and risk when using computer learning methods for stock prediction and trading? High predictive accuracy doesn't always mean high profitability due to costs, volatility, and losses.

Hypothesis: Models with higher predictive accuracy won't always make more money after accounting for transaction costs and other real-world factors. RL methods, which optimize trading strategies directly, may achieve better risk-adjusted returns even if their raw accuracy is lower [6, 5].

2 Summary of Literature Review

The use of machine learning (ML) for stock market prediction has increased substantially in recent years. Patel et al. (2015) demonstrated that combining multiple supervised models, support vector machines (SVM), random forests, and artificial neural networks (ANN), in an ensemble improved directional accuracy compared to individual models [10]. Zhong and Enke (2019) applied principal component analysis (PCA) to reduce noise before training neural networks, improving daily stock return predictions [11]. Nabipour et al. (2020) compared classical ML approaches, such as decision trees and k-nearest neighbors, with deep learning methods including ANN, RNN, and LSTM, finding that deep models performed better on sequential financial data [8]. Parray et al. (2020) provided additional baselines, comparing logistic regression, perceptrons, and SVMs on stock price movements [9].

Although ML can achieve high predictive accuracy, translating predictions into profitable trading strategies remains challenging. Avramov et al. (2022) note that transaction costs and market friction often reduce the practical value of ML signals [1]. Huang et al. (2021) found that models relying solely on technical indicators may overlook fundamental factors such as earnings or book-to-market ratios, suggesting that combining technical and fundamental features can enhance long-term prediction performance [3].

3 Dataset

We will use the Stock Market Data (NASDAQ, NYSE, S&P 500) dataset from Kaggle [7], which provides daily stock prices (open, high, low, close) and volume for thousands of stocks. We will combine all company-level files to create one dataset per market. The data will be cleaned to remove missing values, normalized, and aligned by date. Technical indicators like moving averages, relative strength index (RSI), and volatility measures will be calculated for each stock. Finally, the dataset will be split into training, validation, and test sets.

4 EDA and Methodological Approach

We will examine the dataset for missing values, inconsistencies, and outliers. Summary statistics (mean, standard deviation, min, max) will be calculated for key variables such as daily returns and trading volume. Time-series patterns will be visualized in Tableau for selected stocks and indices using line plots, moving averages, and volatility measures. Correlations among technical indicators and returns will be analyzed in SAS to identify potential predictive features. This ensures data quality and informs feature selection and model design.

Our analysis will compare supervised machine learning (ML) methods—such as linear regression, random forests, and LSTMs—with reinforcement learning (RL) approaches, including deep Q-networks and policy gradient methods. Supervised models will predict future returns or directional movement, while RL agents will learn trading strategies that optimize cumulative reward, accounting for both profit and risk. All models will be trained, validated, and tested on chronologically split datasets for NASDAQ, NYSE, and S&P 500 stocks. Performance metrics will include predictive accuracy, realized profitability, and risk-adjusted measures such as Sharpe ratio and maximum drawdown. Feature engineering will include technical indicators (e.g., moving averages, RSI) and potentially fundamental variables.

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